

elastic **Agent Builder:**
Providing Relevant
Context for Data Driven
Agents



elastic

| The Search
AI Company





**Anderson
Queiroz**
(he / him)
Senior Software
Engineer



Agenda

01	What are Agents?
02	Why do we need Context Engineering?
03	Elastic Agent Builder
04	Demo
05	Resources
06	Q&A

What are Agents?

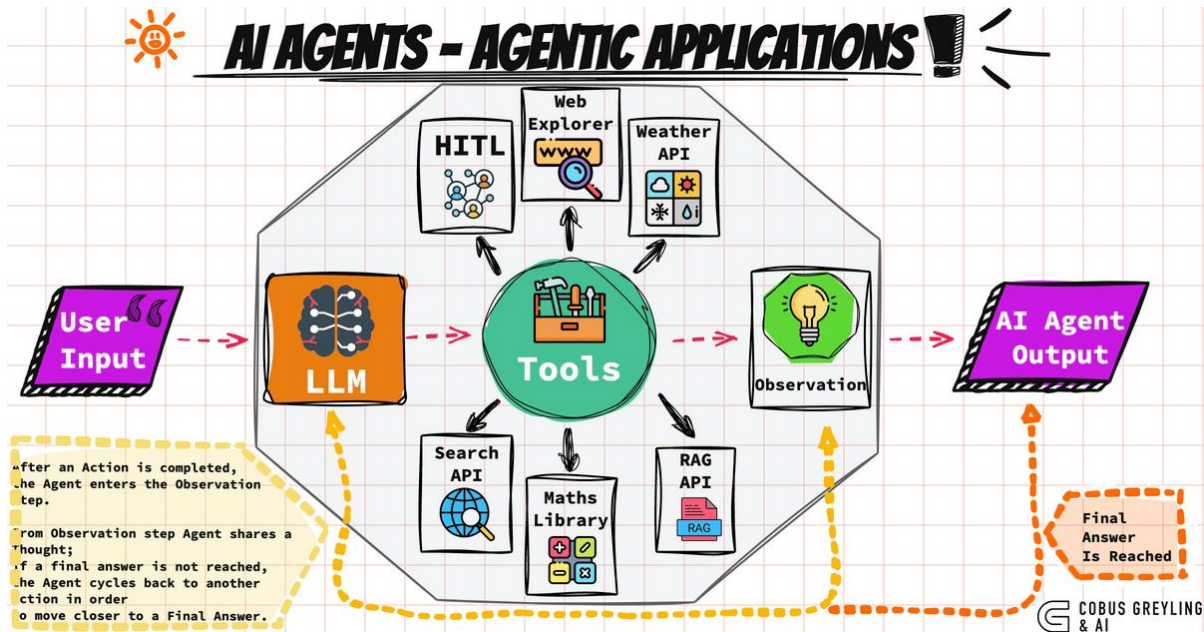
Not a secret agent!



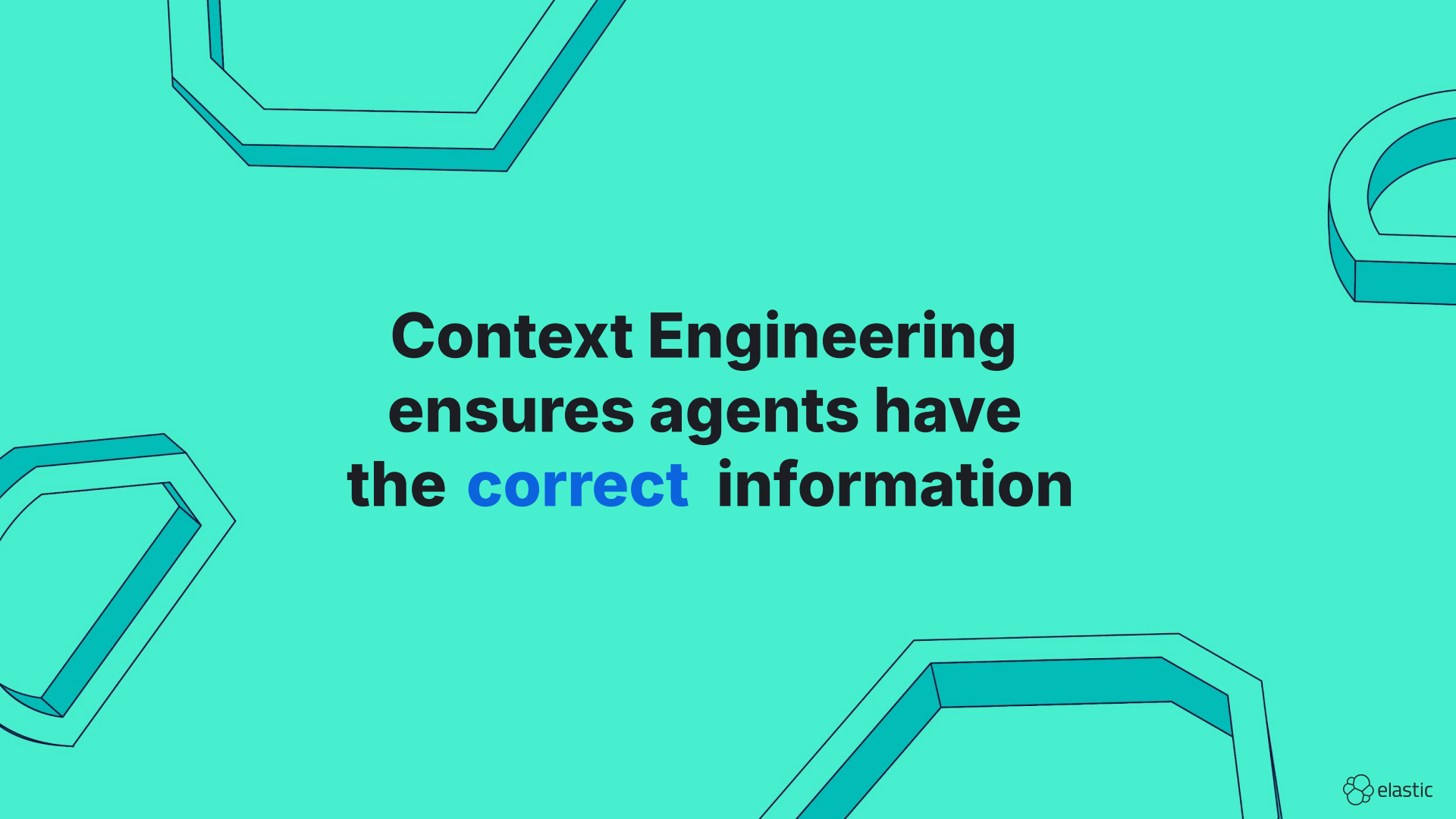
What is an Agent?

An AI agent is a software program that can autonomously perform complex tasks by making decisions, learning from its environment, and using tools to achieve goals

"LLMs autonomously using tools in a loop"



<https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents-definition>



**Context Engineering
ensures agents have
the **correct** information**

Context Window

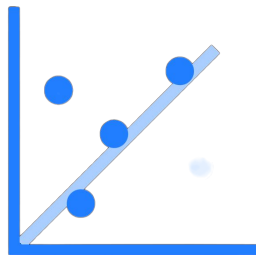
The maximum number of tokens an LLM can process at once.



Why do LLMs Hallucinate?



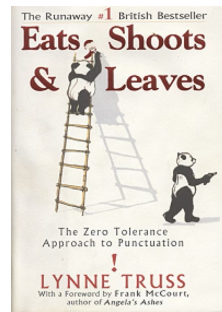
Frozen or Limited Knowledge



Model Overfitting



Dataset Biases



Language Ambiguity

Why Language Models Hallucinate

Adam Tauman Kalai*
OpenAI

Ofir Nachum
OpenAI

Santosh S. Vempala†
Georgia Tech

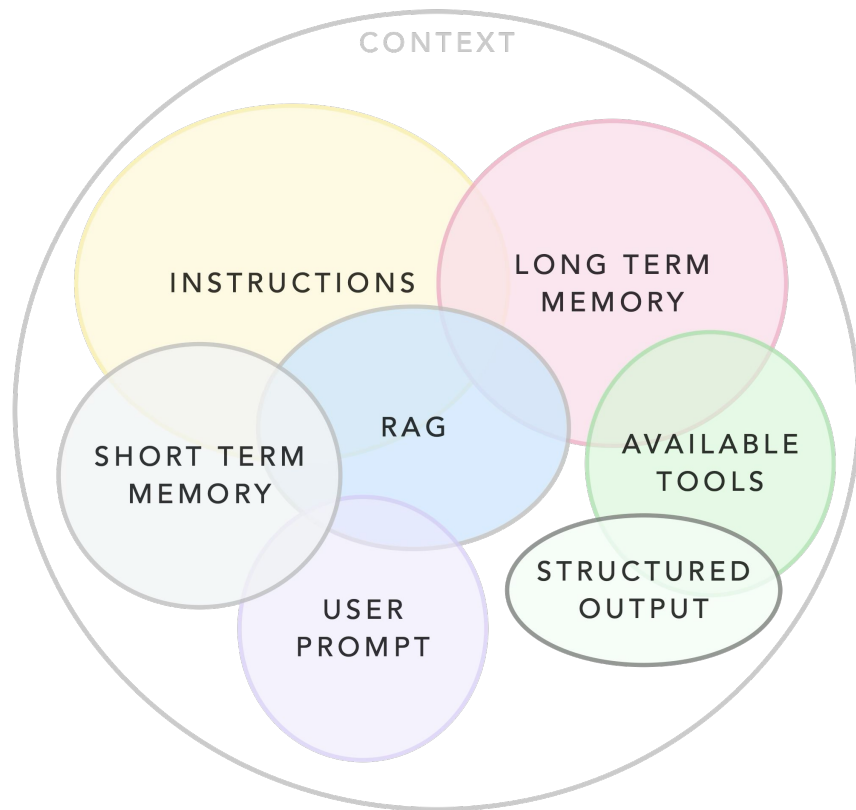
Edwin Zhang
OpenAI

September 4, 2025

Abstract

Like students facing hard exam questions, large language models sometimes guess when uncertain, producing plausible yet incorrect statements instead of admitting uncertainty. Such “hallucinations” persist even in state-of-the-art systems and undermine trust. We argue that language models hallucinate because the training and evaluation procedures reward guessing over acknowledging uncertainty and we analyze the statistical causes of hallucinations in the modern training pipeline. Hallucinations need not be mysterious—they originate simply as errors in binary classification. If incorrect statements cannot be distinguished from facts, then hallucinations in pretrained language models will arise through natural statistical pressures. We then argue that hallucinations persist due to the way most evaluations are graded—language models are optimized to be good test-takers, and guessing when uncertain improves test performance. This “epidemic” of penalizing uncertain responses can only be addressed through a socio-technical mitigation: modifying the scoring of existing benchmarks that are misaligned but dominate leaderboards, rather than introducing additional hallucination evaluations. This change may steer the field toward more trustworthy AI systems.

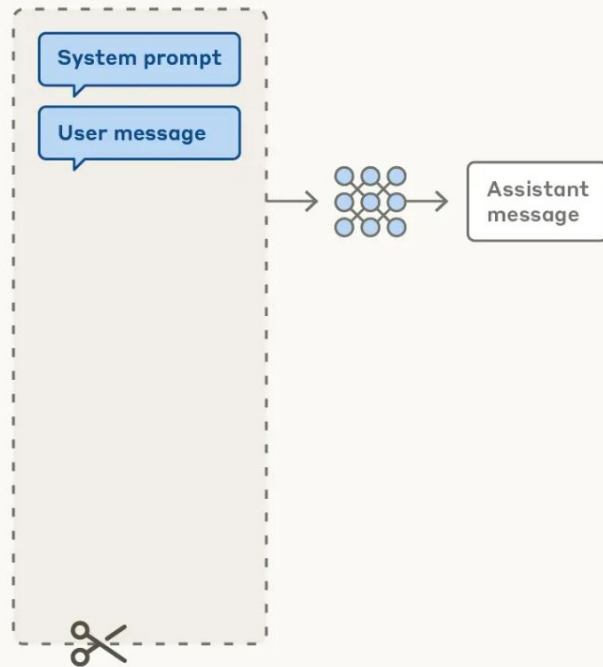
What is Context Engineering?



Prompt engineering vs. context engineering

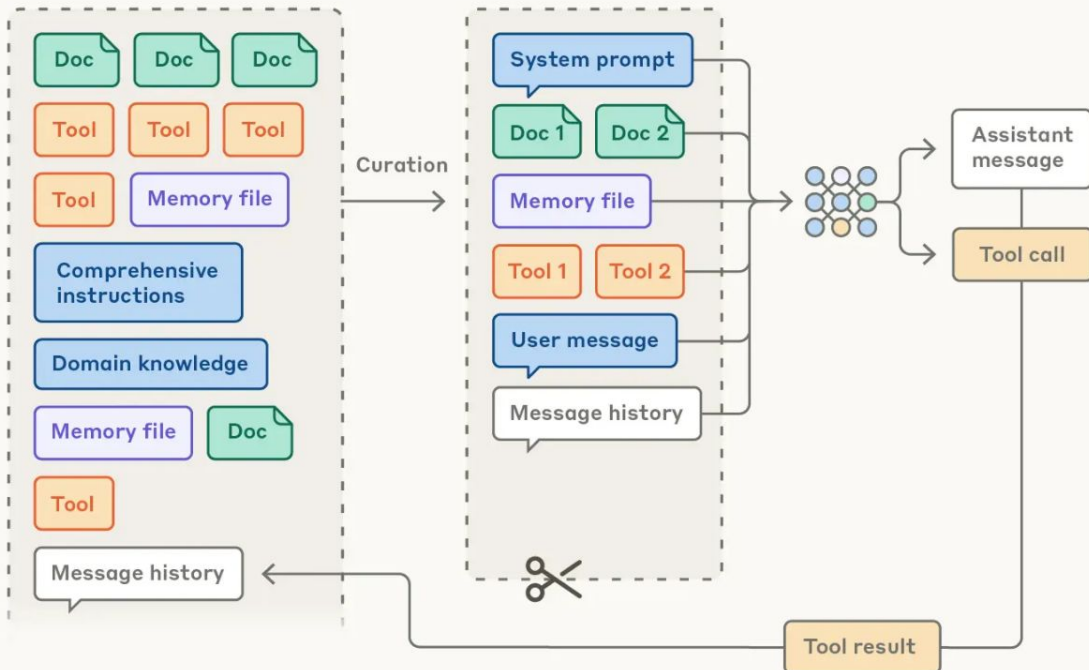
Prompt engineering for single turn queries

Context window



Context engineering for agents

Possible context to give model



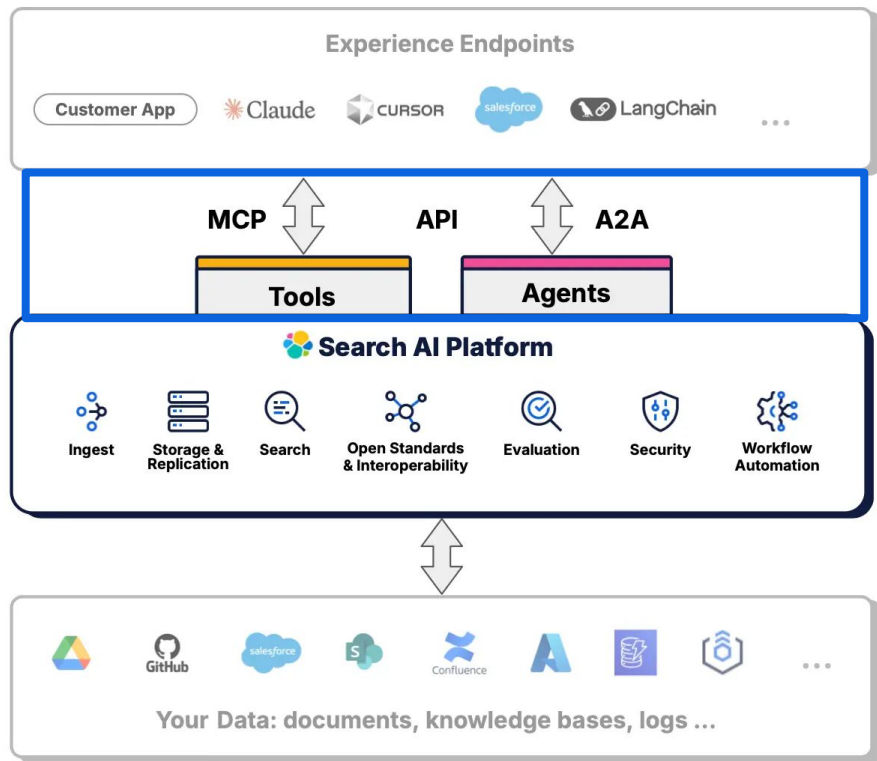
[Credit: Effective context engineering for AI agents | Anthropic](#)



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Agent Builder **allows building of** **data-centric contextual** **agents**

Where does Agent Builder fit?



Capabilities

Elastic Agent Builder has the following features



Chat with your data using the native conversational agent and see full tracing of the steps and results



Leverage built-in tools to find relevant indices, understand data structure and translate native language into structured semantic, lexical or hybrid queries



Build custom tools using ES|QL for use in your own agents

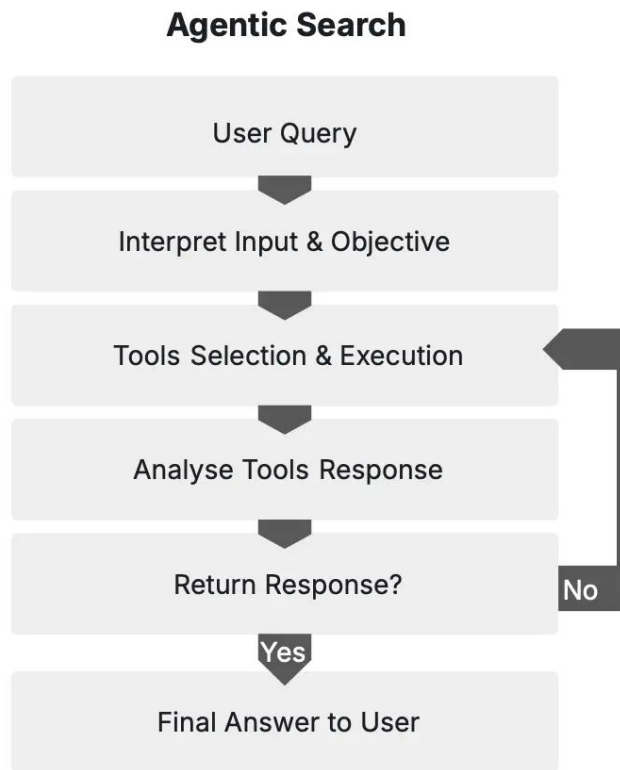


Built in visualizations for chat with editable chart types and the ability to add to Dashboards



Integrate with external agents and apps using MCP and A2A

How do Agents work in Elasticsearch?

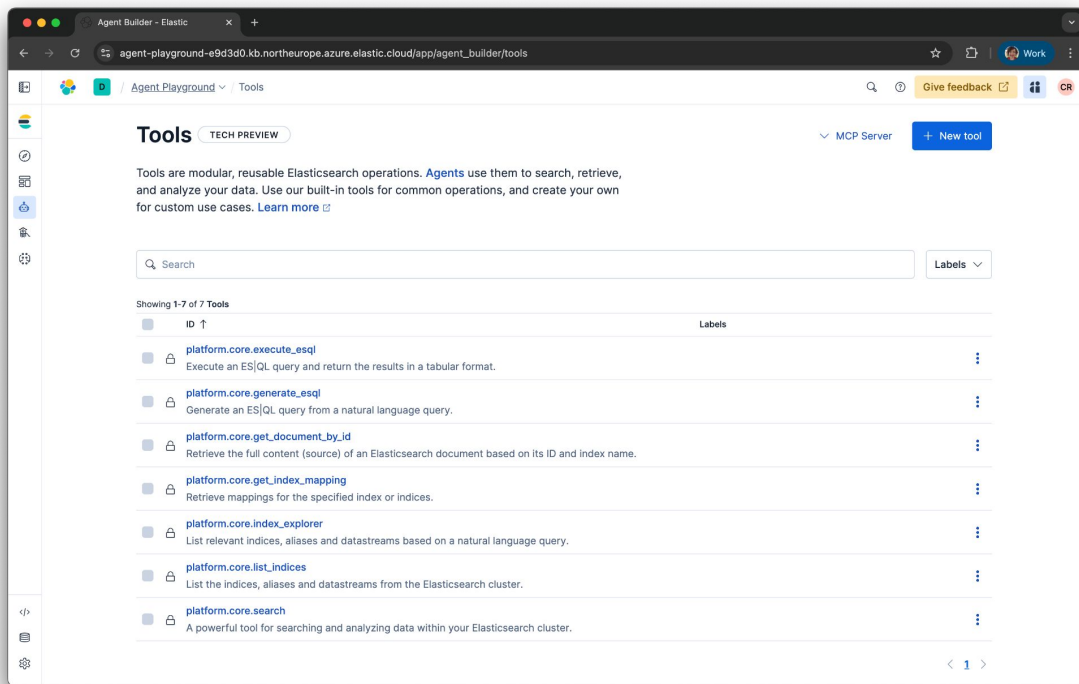


In-built tools

There are currently 7 tools available, prefixed by

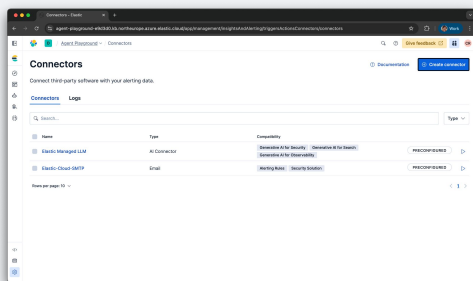
`platform.core`:

- Execute ES|QL queries
- Generate ES|QL query from natural language
- Get full document content based in ID
- Retrieve mappings for one or more specified indices
- List relevant indices, aliases and streams based on natural language query
- List indices, alises and datastreams in the cluster
- Key search tool



Which LLM?

Default Elastic Managed LLM via Elastic Inference Service (EIS)



Connector Create a custom connector using the AI Connector (tech preview)

AI Connector TECHNICAL PREVIEW

Send requests to AI providers such as Amazon Bedrock, OpenAI and more.

Compatibility: Generative AI for Security Generative AI for Search Generative AI for Observability

Connector name

Service

Settings
Model ID

The name of the model to use for the inference task.
[More options](#)

Authentication
API Key

The OpenAI API authentication key. For more details about generating OpenAI API keys, refer to the <https://platform.openai.com/account/api-keys>.

You will need to re-enter your API Key each time you edit the inference endpoint

[Back](#)

Demo

Resources

- [Introducing Elastic's Agent Builder](#)
- [Building AI Agentic workflows with Elasticsearch](#)
- [Your first Elastic Agent: From a single query to an AI-powered chat](#)
- [Connecting Elastic Agents to Gemini Enterprise via A2A protocol](#)

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With Elasticsearch's [most-downloaded vector database](#), ideas become code – build configurable search and trusted GenAI experiences that scale from prototype to production fast.

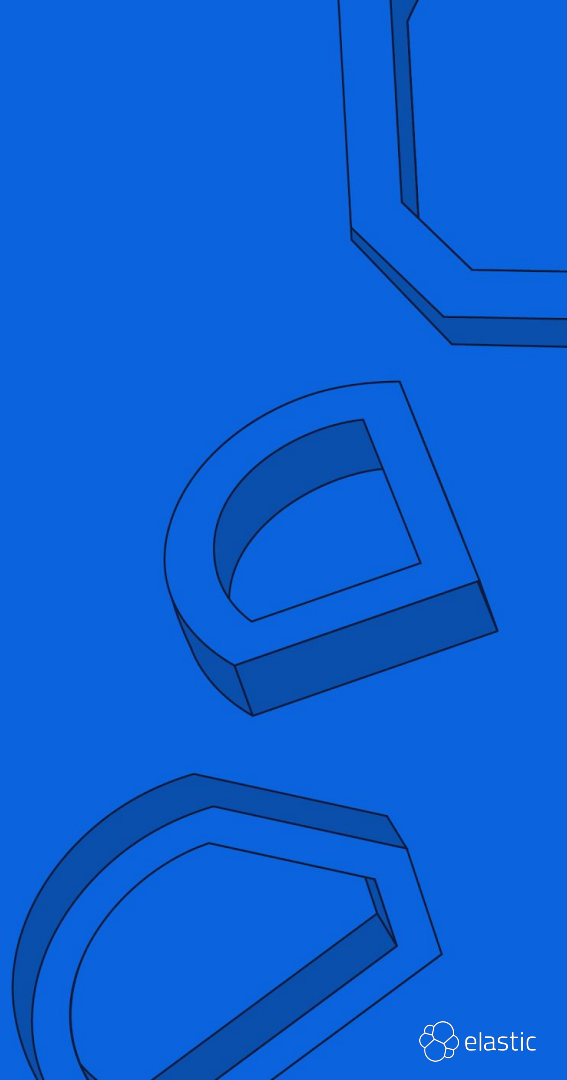
Elasticsearch is where pilots roll into production flawlessly without a steep learning curve. Welcome to rolling out new indices, models, and search techniques in a flash. Trusted by millions for its speed, scale, and relevance — Elasticsearch gives you a decade-long head start with a flexible AI search toolkit and total provisioning flexibility.

[Basics](#)

October 29, 2025

How to deploy Elasticsearch on an Azure Virtual Machine

Questions?



Questions?

Thank you!



Slides/Demo:

<https://github.com/elastic/meetups/tree/main/Berlin>



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LinkedIn: **Anderson Queiroz**
GitHub: **@AndersonQ**
anderson.queiroz@elastic.co